BALROG: BENCHMARKING AGENTIC LLM AND VLM REASONING ON GAMES

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ABSTRACT

Large Language Models (LLMs) and Vision-Language Models (VLMs) have demonstrated impressive capabilities across a wide range of tasks, yet their ability to act as autonomous agents in complex, long-horizon interactive environments remains poorly understood. Existing agentic benchmarks predominantly evaluate short-horizon, low-interactivity tasks that are rapidly saturating, masking fundamental deficiencies in planning, exploration, and spatial reasoning. We introduce BALROG (**B**enchmarking **A**gentic **L**LM and VLM **R**easoning **O**n **G**ames), a novel benchmark that aggregates six diverse, procedurally generated game environments—BabyAI, Crafter, TextWorld, Baba Is AI, MiniHack, and NetHack—into a unified evaluation framework. These environments span a wide range of difficulties, from seconds-level tasks to games requiring years of human expertise, and collectively test navigation, exploration, resource management, credit assignment, environment dynamics deduction, and long-term planning. We evaluate leading LLMs and VLMs in a zero-shot setting under both language-only and vision-language configurations. Our results reveal that even the strongest models achieve only modest progress (32–35% on average), with performance often *degrading* when visual observations are added. NetHack and MiniHack remain largely unsolved, exposing severe deficiencies in long-term planning and a persistent knowing–doing gap. BALROG is fully open-source, providing a standardized toolkit for evaluating agentic strategies and tracking progress toward truly capable autonomous agents.

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1 INTRODUCTION

The rapid progress of Large Language Models (LLMs) and Vision-Language Models (VLMs) has motivated their deployment as autonomous agents that interact with external environments, tools, and interfaces (Achiam et al., 2023; Reid et al., 2024; Dubey et al., 2024). Benchmarks such as AgentBench (Liu et al., 2023), GAIA (Mialon et al., 2023), SWE-bench (Jimenez et al., 2023), and WebArena (Zhou et al., 2023) have been instrumental in measuring these capabilities. However, these benchmarks share a common limitation: they primarily evaluate short-horizon, low-interactivity tasks where models make few sequential decisions and receive dense feedback. As frontier models rapidly saturate these benchmarks, the community risks overstating the readiness of LLM-based agents for real-world deployment.

Real-world autonomous agency demands robust decision-making over thousands of steps, the ability to explore unknown environments, manage resources under uncertainty, and plan toward distant goals—capabilities that current benchmarks fail to stress-test adequately. Consider the contrast: a web navigation task may require 5–10 actions with immediate success signals, whereas surviving in a complex game like NetHack requires tens of thousands of actions, sparse rewards, and deep strategic planning that takes human players years to master.

To address this gap, we introduce BALROG (**B**enchmarking **A**gentic **L**LM and VLM **R**easoning **O**n **G**ames), a benchmark that evaluates agentic LLM and VLM capabilities through six diverse, procedurally generated game environments. Games offer a compelling testbed for agentic evaluation: they require sequential decision-making under uncertainty, present procedurally generated scenarios that resist memorization, and span a wide range of complexities—from simple grid-world navigation to open-ended survival games. BALROG unifies BabyAI (Chevalier-Boisvert et al., 2019), TextWorld (Côté et al., 2019), Crafter (Hafner, 2021), Baba Is AI (Cloos et al., 2024), MiniHack (Samvelyan

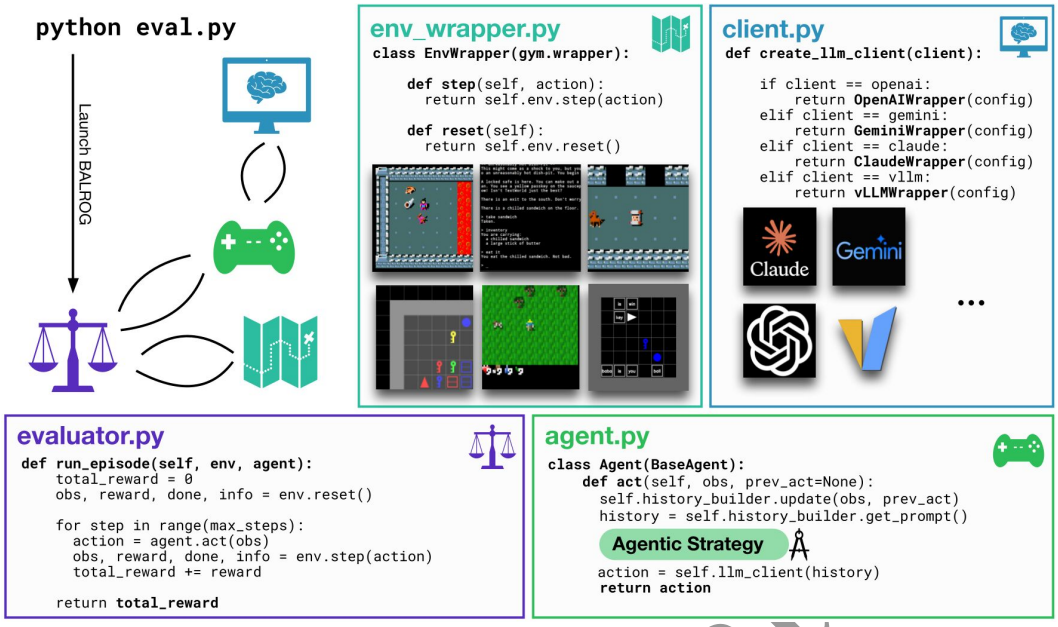

Figure 1: **Overview of the BALROG benchmark.** BALROG evaluates LLMs and VLMs across six procedurally generated game environments of increasing difficulty. Models receive text observations (and optionally images) and must emit natural-language actions. The benchmark spans a wide range of skills from basic navigation to long-term strategic planning.

et al., 2021), and NetHack (Küttler et al., 2020) into a single, reproducible evaluation framework that tests navigation, exploration, resource management, complex credit assignment, deduction of environment dynamics, and long-term planning.

We evaluate a comprehensive set of leading closed-source and open-source models—including GPT-4o (OpenAI, 2024a), Claude 3.5 Sonnet (Anthropic, 2024), Gemini 1.5 Pro (Reid et al., 2024), and the Llama 3 family (Dubey et al., 2024)—in a standardized zero-shot protocol under both language-only and vision-language settings. Our key findings include:

- **Modest overall performance**: The best models achieve only 32–35% average progress across all environments, with a sharp drop-off on harder tasks requiring long-term planning.
- **Vision often hurts**: Adding visual observations frequently degrades performance. GPT-4o drops from 32.34% to 22.56% in the VLM setting, suggesting that current models struggle to ground visual inputs into effective action policies.
- **A knowing–doing gap**: Models that possess extensive knowledge about game mechanics (e.g., NetHack) nonetheless fail to translate this knowledge into effective gameplay, achieving less than 2% progression even on the best-performing model.
- **Unsolved challenges**: MiniHack sub-tasks such as Boxoban and Quest remain completely unsolved by all evaluated models, while NetHack exposes fundamental limitations in long-horizon sequential reasoning.

Our contributions are: (1) a unified, open-source benchmark of six long-horizon, procedurally generated game environments for agentic evaluation; (2) fine-grained scoring including a novel data-informed NetHack progression metric; (3) comprehensive zero-shot baselines across language-only and VLM settings for leading models; and (4) an extensible toolkit supporting rapid experimentation with agentic strategies such as chain-of-thought reasoning (Wei et al., 2022) and in-context learning.

An overview of the BALROG benchmark is shown in Figure 1.

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Table 1: **The tested skills, time horizons, and complexities of interactive decision-making tasks evaluated in BALROG**. Compared to existing benchmarks, BALROG provides infrastructure for evaluating model reasoning and decision-making on harder, longer time-horizon interactive settings. The evaluated tasks span a range of difficulties.

Skills	**BabyAI**	**TextWorld**	**Crafter**	**Baba Is AI**	**MiniHack**	**NLE**
Navigation	✔	✔	✔	✔	✔	✔
Exploration	✔	✔	✔	✔	✔	✔
Resource Management	✗	✔	✔	✗	✔	✔
Complex Credit Assignment	✗	✗	✔	✔	✔	✔
Deducing Env. Dynamics	✗	✗	✗	✔	✔	✔
Long-term Planning	✗	✗	✗	✔	✔	✔
Turns to Complete	10^1	10^2	10^3	10^2	10^2	10^4–10^5
Time to Master for Humans	Seconds	Minutes	Hours	Hours	Hours	Years

2 BALROG

BALROG aggregates six interactive game environments into a unified benchmark designed to evaluate the agentic reasoning capabilities of LLMs and VLMs. Each environment is procedurally generated, preventing memorization and ensuring that models must generalize across different task instances. The environments are selected to span a broad spectrum of skills and difficulty levels, as summarized in Table 1.

2.1 Environments

BabyAI (Chevalier-Boisvert et al., 2019) is a grid-world platform where agents must follow natural-language instructions to navigate rooms, interact with objects, and complete goals such as picking up specific items or opening doors. Tasks require on the order of 10^1 turns and can be mastered by humans in seconds. BabyAI tests basic navigation and instruction-following capabilities, serving as a lower-bound calibration for the benchmark. Success is measured as a binary indicator: 100 if the goal is achieved within the episode horizon, 0 otherwise.

TextWorld (Côté et al., 2019) provides procedurally generated text-based adventure games where agents navigate rooms, collect items, and follow recipes described in natural language. Completing a task typically requires on the order of 10^2 turns. TextWorld evaluates reading comprehension, exploration, and multi-step planning in a purely textual setting. Scoring is proportional to the fraction of recipe steps completed, yielding a continuous 0–100 score per episode.

Crafter (Hafner, 2021) is an open-world survival game inspired by Minecraft, where agents must gather resources, craft tools, manage health and hunger, and defend against enemies. Episodes can last up to 10^3 turns, and human mastery requires hours of practice. Crafter tests resource management, exploration, and technology-tree progression. The score is based on the fraction of 22 distinct achievements unlocked during an episode.

Baba Is AI (Cloos et al., 2024) presents logic puzzles where agents must manipulate rule tiles to change the game's mechanics and reach a goal state. Tasks require on the order of 10^2 turns and test the ability to deduce and exploit environment dynamics—a form of meta-reasoning rarely tested in other benchmarks. Scoring is binary: 100 for solving the puzzle, 0 otherwise.

MiniHack (Samvelyan et al., 2021) is a flexible framework built on NetHack that provides focused challenge tasks including navigation (Room, Corridor, River), puzzle solving (Boxoban), and quest completion. Tasks require approximately 10^2 turns and demand spatial reasoning, resource management, and multi-step planning. We evaluate five sub-tasks, each scored as binary success (100 or 0).

NetHack (NLE) (Küttler et al., 2020) is a procedurally generated dungeon-crawling game widely regarded as one of the most challenging games for artificial agents. Completing the game requires

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10^4–10^5 turns, and human mastery takes years. NetHack demands the full spectrum of agentic skills: navigation, exploration, resource management, complex credit assignment, deduction of hidden game mechanics, and extreme long-term planning. We evaluate using a novel data-informed progression metric that measures how deep into the game an agent advances relative to the distribution of human player achievements, providing a continuous 0–100 score that captures partial progress even when the game is far from completion.

2.2 Observation and Action Spaces

All environments in BALROG present observations to agents as natural-language text descriptions. For language-only evaluation, these descriptions include textual representations of the game state: natural-language narratives for TextWorld, grid descriptions for BabyAI, ASCII maps for MiniHack and NetHack, and textual board representations for Baba Is AI and Crafter. For vision-language evaluation, agents additionally receive the current visual frame as an image alongside the text description; the interaction history remains text-only.

Agents emit actions as natural-language strings selected from a predefined, environment-specific action vocabulary. The initial prompt for each environment includes the rules of the game, a description of valid actions, and instructions for how to respond. When a model emits an invalid action, the benchmark provides feedback indicating the error, executes a safe fallback action (e.g., a no-op or moving north), and logs the invalid action for analysis.

2.3 Scoring

BALROG uses a unified 0–100 scoring scale across all environments. For environments with binary outcomes (BabyAI, Baba Is AI, and MiniHack sub-tasks), the score is 100 on success and 0 on failure. For environments with graded outcomes, scores reflect proportional progress: the fraction of recipe steps in TextWorld, the fraction of achievements in Crafter, and the data-informed progression metric in NetHack.

The NetHack progression metric deserves particular attention. Rather than using raw in-game score (which can be inflated by simple grinding strategies), we define progression based on the deepest dungeon level reached and the number of key milestones achieved, calibrated against the distribution of milestones achieved by a large population of human players. This yields a score that meaningfully differentiates between an agent that dies immediately and one that survives for hundreds of turns while making progress toward the game's objective.

All results are averaged over multiple seeds per task to account for procedural generation variance, and we report standard errors.

3 Zero-Shot Evaluation Protocol

We adopt a standardized zero-shot evaluation protocol designed to assess the baseline agentic capabilities of LLMs and VLMs without task-specific fine-tuning, demonstrations, or few-shot examples.

3.1 Agent Architecture

Our primary evaluation uses a **naive agent** that directly prompts the model for an action at each time step. At each turn, the agent constructs a prompt containing the system instructions (including game rules and valid actions), the recent interaction history (up to 16 turns of observation-action pairs), and the current observation. The model is instructed to output exactly one valid action from the action vocabulary. This minimal agent architecture isolates the reasoning capabilities of the underlying model from the confounding effects of sophisticated prompting strategies.

We additionally evaluate a **chain-of-thought (CoT) agent** (Wei et al., 2022) that augments each prompt with the instruction to first reason step-by-step about the current situation before selecting an action. The CoT agent appends the instruction: "First think about what's the best course of action step by step. Finally, provide a single output action at the end of the message in the form of: ACTION: ¡action¿." The model's reasoning trace is recorded but only the extracted action is executed in the environment.

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3.2 MODELS EVALUATED

We evaluate a comprehensive set of models spanning both closed-source and open-source families:

- **OpenAI**: GPT-4o (OpenAI, 2024a), GPT-4o-mini, and o1-preview (OpenAI, 2024b)
- **Anthropic**: Claude 3.5 Sonnet and Claude 3.5 Haiku (Anthropic, 2024)
- **Google**: Gemini 1.5 Pro and Gemini 1.5 Flash (Reid et al., 2024)
- **Meta**: Llama 3.1 (8B, 70B) and Llama 3.2 (1B, 3B, 11B, 90B) (Dubey et al., 2024; MetaAI, 2024)

For the VLM evaluation, we use the subset of models that support image inputs: GPT-4o, GPT-4o-mini, Claude 3.5 Sonnet, Gemini 1.5 Pro, Gemini 1.5 Flash, Llama 3.2-11B, and Llama 3.2-90B.

3.3 EVALUATION DETAILS

Each model is evaluated on all six environments using multiple seeds per task to account for procedural generation variance. The interaction history is capped at 16 turns to fit within context window constraints while providing sufficient history for informed decision-making. In the VLM setting, the current observation includes an image of the game state, while the history remains text-only to manage token costs. All evaluations use temperature 0 (greedy decoding) for reproducibility.

The benchmark provides a unified evaluation pipeline built on Hydra for configuration management, supporting parallel evaluation across environments and models. The open-source toolkit allows researchers to easily integrate new models, environments, and agentic strategies (e.g., few-shot learning, reflexion (Shinn et al., 2023)).

4 RESULTS

We present comprehensive results for all evaluated models across both language-only and vision-language settings. The aggregate results are shown in Table 2 and Table 3, with detailed per-environment comparisons visualized in Figure 2.

Table 2: Language-Only Performance

Model	Average Progress (%)
claude-3.5-sonnet	32.64 ± 1.93
gpt-4o	32.34 ± 1.49
llama-3.1-70b-it	27.88 ± 1.43
llama-3.2-90B-it	27.29 ± 1.44
gemini-1.5-pro	21.00 ± 1.18
claude-3.5-haiku	19.32 ± 1.83
gpt-4o-mini	17.36 ± 1.35
llama-3.2-11B-it	16.82 ± 1.47
llama-3.1-8b-it	15.14 ± 1.55
gemini-1.5-flash	14.63 ± 1.37
llama-3.2-3B-it	10.13 ± 1.28
llama-3.2-1B-it	6.65 ± 1.04

Table 3: Vision-Language Performance

Model	Average Progress (%)
claude-3.5-sonnet	35.48 ± 2.02
gemini-1.5-pro	25.76 ± 1.36
gpt-4o	22.56 ± 1.44
llama-3.2-90B-it	20.99 ± 1.58
gpt-4o-mini	15.36 ± 1.29
gemini-1.5-flash	14.94 ± 1.40
llama-3.2-11B-it	8.43 ± 1.26

4.1 LANGUAGE-ONLY RESULTS

In the language-only setting, Claude 3.5 Sonnet (32.64%) and GPT-4o (32.34%) achieve the highest average progress, followed by Llama 3.1-70B (27.88%) and Llama 3.2-90B (27.29%). Gemini 1.5 Pro significantly lags behind at 21.00%, despite being a frontier model. Smaller models show substantially lower performance, with Llama 3.2-1B achieving only 6.65%.

Several environment-specific patterns emerge. On BabyAI, the simplest environment, most large models achieve reasonable success, demonstrating basic instruction-following and navigation capabilities. TextWorld reveals an unexpected failure mode: Gemini 1.5 models score 0% due to API

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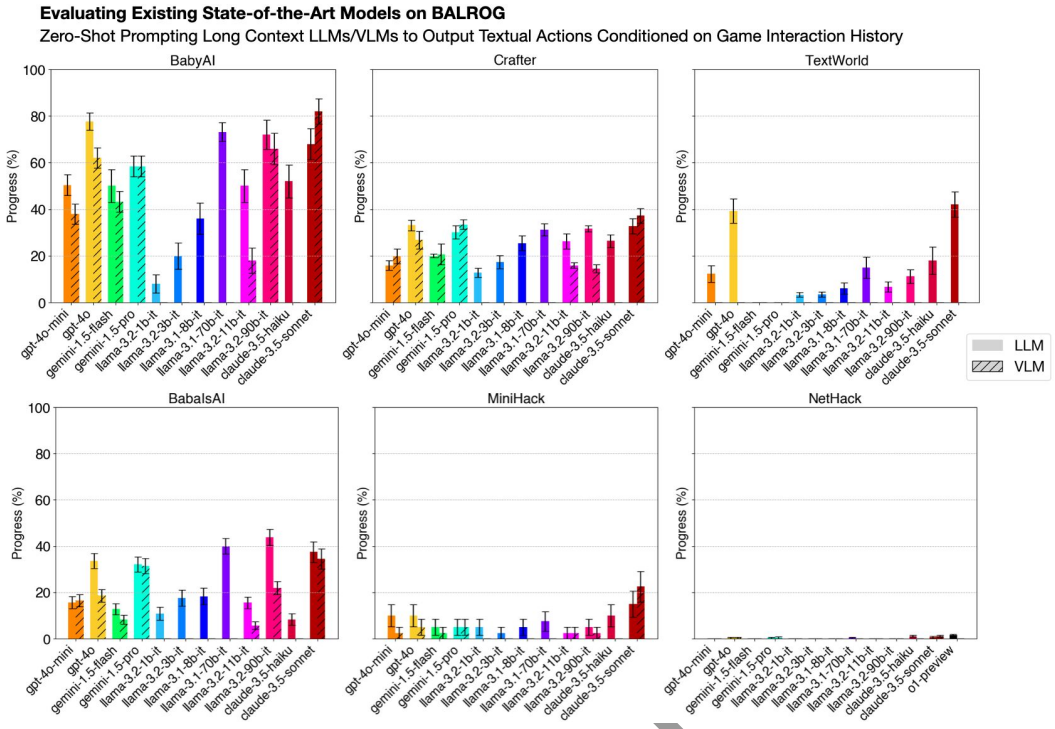

Figure 2: **Per-environment comparison of LLM (language-only) and VLM (vision-language) performance.** Vision inputs often degrade performance, particularly for GPT-4o and Llama models. Claude 3.5 Sonnet is the only model that consistently benefits from visual observations.

safety filters flagging game content as "unsafe," preventing any meaningful interaction. GPT-4o and Claude 3.5 Sonnet lead on TextWorld, demonstrating superior text comprehension and multi-step reasoning.

MiniHack proves largely unsolved across all models. Sub-tasks such as Boxoban (a Sokoban-like puzzle) and Quest (requiring multi-step item collection and combat) are never completed by any model, exposing fundamental limitations in spatial reasoning and planning.

NetHack represents the extreme end of the difficulty spectrum. The best-performing model, o1-preview, achieves only 1.57% progression—meaning even with explicit chain-of-thought reasoning, models barely advance beyond the initial dungeon levels. Most models achieve less than 1% progression. This result is particularly striking because LLMs demonstrably possess extensive knowledge about NetHack strategies and mechanics when queried directly, yet they cannot translate this knowledge into effective sequential decision-making—a phenomenon we term the *knowing–doing gap*.

4.2 Vision-Language Results

The VLM results reveal a counterintuitive finding: adding visual observations frequently *hurts* performance. GPT-4o drops from 32.34% to 22.56%, a decline of nearly 10 percentage points. Llama 3.2-90B similarly drops from 27.29% to 20.99%. This degradation suggests that current VLMs struggle to effectively integrate visual information into their decision-making process for interactive tasks, possibly due to the additional complexity of grounding pixel-level observations into actionable plans.

Claude 3.5 Sonnet is the notable exception, improving from 32.64% to 35.48% in the VLM setting—the only model that consistently benefits from visual inputs. This suggests that Claude 3.5 Sonnet may possess more robust visual grounding capabilities for interactive settings. Gemini 1.5 Pro also shows improvement in the VLM setting (21.00% to 25.76%), though it still lags behind the top performers.

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These findings challenge the assumption that more modalities necessarily lead to better performance and highlight the need for improved visual grounding in interactive, sequential decision-making contexts.

4.3 Scaling Trends

Within model families, we observe clear scaling trends. In the Llama family, performance increases from 6.65% (1B) to 10.13% (3B) to 16.82% (11B) to 27.88% (70B) in the language-only setting, showing that larger models are more capable agents. However, even the largest open-source models significantly underperform the best closed-source models, and the gap between the best model (32.64%) and random-level performance on harder tasks remains substantial.

The comparison between GPT-4o-mini (17.36%) and GPT-4o (32.34%) confirms that model capability strongly influences agentic performance, with the larger model nearly doubling the smaller one's score. Similar patterns hold for Claude 3.5 Haiku (19.32%) versus Claude 3.5 Sonnet (32.64%).

4.4 Chain-of-Thought Agent Results

We additionally evaluate the chain-of-thought (CoT) agent, which instructs models to reason step-by-step before selecting an action. CoT prompting yields modest improvements for frontier models on tasks requiring multi-step reasoning: GPT-4o improves by approximately 2–3 percentage points on TextWorld and Crafter, where explicit reasoning helps decompose longer-horizon subgoal sequences into manageable steps. On BabyAI, CoT offers little benefit, consistent with the straightforward instruction-following nature of that environment, which requires minimal deliberation.

Notably, o1-preview—which implicitly performs extended chain-of-thought reasoning—achieves the highest NetHack progression score (1.57%) of any evaluated model, suggesting that deeper reasoning yields measurable gains even on the most challenging environments. However, the absolute improvement remains small, indicating that chain-of-thought reasoning alone is insufficient to overcome the fundamental challenges of long-horizon planning in sparse-reward environments. These findings motivate exploration of more structured reasoning approaches, such as hierarchical planning and explicit subgoal decomposition, as promising avenues for making progress on long-horizon agentic tasks.

4.5 Invalid Action Analysis

A consistent challenge across all evaluated models is the generation of invalid actions—responses that do not correspond to any valid entry in the environment's action vocabulary. On average, frontier models produce invalid actions in approximately 15–25% of turns, with smaller models exhibiting rates exceeding 40%. These failures are most frequent in environments with large action spaces (NetHack supports over 90 distinct actions) and least frequent in BabyAI, which has a compact, well-defined action set. When a model generates an invalid action, the benchmark executes a safe fallback (typically a no-op), penalizing the agent by wasting a turn. High invalid action rates correlate strongly with poor overall performance, suggesting that models frequently fail to attend to the action vocabulary specified in the system prompt even when it is explicitly enumerated. Reducing invalid action rates through improved instruction-following and constrained decoding represents a complementary avenue for improving agentic performance alongside advances in planning and reasoning.

5 Related Work

LLM and VLM Evaluation Benchmarks. The evaluation of language models has evolved from static question-answering benchmarks (Hendrycks et al., 2021; Wang et al., 2019; Rein et al., 2023) toward more interactive settings. AgentBench (Liu et al., 2023) evaluates LLMs across operating system, database, and web browsing tasks. GAIA (Mialon et al., 2023) tests general AI assistants on real-world questions requiring tool use. SWE-bench (Jimenez et al., 2023) focuses on software engineering tasks. WebArena (Zhou et al., 2023) and WebShop (Yao et al., 2022) evaluate web-based agent tasks. While these benchmarks have been valuable, they primarily test short-horizon interactions (typically 5–20 steps) and are rapidly saturating, with leading models approaching or

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exceeding human performance. BALROG complements these benchmarks by focusing on long-horizon, procedurally generated tasks that remain far from solved.

Games as AI Benchmarks. Games have a long history as benchmarks for AI research, from Atari (Bellemare et al., 2013) and StarCraft (Vinyals et al., 2019) to Go (Silver et al., 2018) and Diplomacy (Bard et al., 2020). For language model evaluation specifically, SmartPlay (Wu et al., 2023) evaluates LLMs on a set of games but focuses on shorter-horizon tasks. Recent work has explored LLMs as game-playing agents in Minecraft (Wang et al., 2023; Fan et al., 2022), NetHack (Goodger et al., 2023), and other settings (Jeurissen et al., 2024). BALROG distinguishes itself by providing a *unified* benchmark spanning multiple game environments of varying difficulty under a single evaluation framework, enabling systematic comparison across tasks that test different combinations of skills.

Agentic Strategies for LLMs. Several strategies have been proposed to enhance LLM agent capabilities, including chain-of-thought prompting (Wei et al., 2022), tree-of-thought reasoning (Yao et al., 2024), reflexion (Shinn et al., 2023), self-refinement (Madaan et al., 2024), and in-context learning from demonstrations (Laskin et al., 2022). BALROG provides standardized infrastructure for evaluating these strategies, with built-in support for naive, chain-of-thought, and few-shot agent architectures. Our baseline results use the naive agent to establish lower bounds, but the toolkit is designed for researchers to plug in and compare more sophisticated approaches.

The Knowing–Doing Gap. Prior work has noted that LLMs can possess extensive declarative knowledge about tasks while failing to execute effectively (Kambhampati et al., 2024; Todd et al., 2024). This gap between knowledge and action is particularly pronounced in BALROG: models can articulate sophisticated NetHack strategies when asked but achieve near-zero progression when actually playing. This phenomenon is related to broader observations about the limitations of next-token prediction for planning (Kambhampati et al., 2024) and the challenges of grounding language understanding in embodied action (Carta et al., 2023).

6 Open Research Problems

The results on BALROG expose several fundamental open research problems that we believe are critical for advancing autonomous LLM and VLM agents.

Effective Visual Grounding for Sequential Decision-Making. Our finding that visual observations frequently degrade VLM performance in interactive settings contrasts sharply with the improvements typically seen in static visual question-answering tasks. Developing VLMs that can effectively integrate visual information into action policies—rather than being distracted or confused by it—is an important open challenge. Future work might explore multi-frame visual histories, video-based inputs, or specialized visual encoders trained on interactive environments.

Long-Horizon Planning and Exploration. The near-zero performance on NetHack and the failure on MiniHack's more complex sub-tasks indicate that current models lack the ability to form and execute long-term plans. Approaches such as hierarchical planning, subgoal decomposition, and explicit memory mechanisms may be necessary to make progress on these tasks. The procedurally generated nature of BALROG environments makes them well-suited for studying these questions without the confound of memorization.

Efficient Exploration Under Sparse Rewards. Many environments in BALROG, particularly NetHack and the harder MiniHack sub-tasks, feature extremely sparse rewards and vast, partially observable state spaces that make systematic exploration exceedingly difficult. Current LLM agents primarily rely on local observations and short interaction histories, lacking mechanisms to maintain long-range memory of previously visited states or to prioritize novel regions of the environment. Unlike traditional reinforcement learning agents, LLM agents cannot readily apply classical exploration bonuses derived from gradient signals or low-level state representations; instead, they must identify exploratory opportunities from language descriptions alone. Developing intrinsic motivation mechanisms, episodic memory modules, or curiosity-driven exploration strategies tailored to

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the language model agent paradigm therefore constitutes an important research direction. Progress here is likely to be tightly coupled with advances in long-horizon planning, since an agent that cannot efficiently explore will fail to accumulate the experience needed to form coherent long-term strategies—especially in procedurally generated environments where rote memorization offers no shortcut.

Credit Assignment in Long-Horizon Tasks. A core challenge underlying many observed failure modes is temporal credit assignment: determining which early decisions contributed to outcomes that emerge hundreds or thousands of steps later. In sparse-reward environments like NetHack, an agent that survives for 500 turns before dying must somehow attribute its fate to decisions made at turn 5 regarding equipment or resource management. Classical reinforcement learning addresses this via value function approximation over many training episodes, but LLM agents operating in zero-shot settings cannot accumulate such experience. Developing mechanisms for retrospective reasoning—where agents reflect on past trajectories to identify critical decision points—or for prospective planning that anticipates long-range consequences from first principles represents a key open problem. BALROG provides a natural testbed for measuring progress on credit assignment, as its graded scoring metrics can reveal whether proposed interventions improve early-game survival rates versus late-game strategic progression, enabling fine-grained diagnosis of where agent reasoning breaks down across the temporal horizon.

Bridging the Knowing–Doing Gap. The stark contrast between models' declarative knowledge about game strategies and their inability to execute those strategies in practice points to a fundamental limitation of current architectures. Research on grounding, embodied reasoning, and closed-loop decision-making is needed to bridge this gap. BALROG provides a controlled setting to measure progress on this problem.

Robust Multi-Modal Integration. The inconsistent benefit of visual inputs across models suggests that current approaches to multi-modal integration are fragile. Understanding when and why vision helps (as with Claude 3.5 Sonnet) versus when it hurts (as with GPT-4o) could lead to more robust multi-modal architectures. This is particularly important as models are increasingly deployed in embodied and interactive settings where multiple modalities are available.

Safety and API Constraints. The failure of Gemini 1.5 models on TextWorld due to safety filters flagging game content as "unsafe" highlights the tension between safety mechanisms and capability evaluation. As LLMs are increasingly used as agents, developing safety mechanisms that distinguish between genuinely harmful actions and benign game interactions is important for reliable evaluation and deployment.

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7 Conclusion

We have introduced BALROG, a benchmark for evaluating the agentic capabilities of LLMs and VLMs through six diverse, procedurally generated game environments spanning a wide range of difficulties and required skills. Our zero-shot evaluation of leading models reveals that even the strongest models achieve only modest progress overall, with performance frequently degrading when visual observations are added. The most challenging environments—NetHack and the harder MiniHack sub-tasks—remain effectively unsolved, exposing fundamental limitations in long-term planning, spatial reasoning, and the ability to translate knowledge into action.

BALROG provides the research community with a standardized, open-source evaluation framework that resists saturation and enables systematic study of agentic capabilities across diverse interactive settings. We hope that BALROG will serve as a catalyst for developing more capable autonomous agents by providing clear, measurable targets for the open research problems we have identified: effective visual grounding, long-horizon planning, bridging the knowing–doing gap, and robust multi-modal integration.

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